Summary in Graph

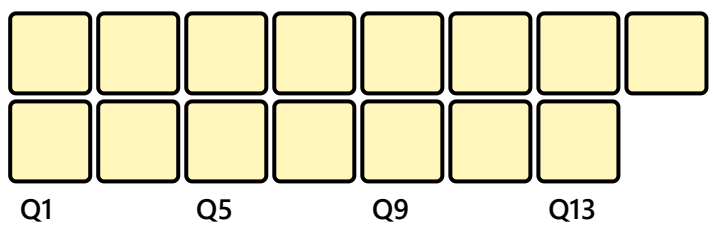
Exam Summary (GO Classes CS Test Series 2026 | Theory of Computation | Topic Wise Test 5)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	15 (5 + 10)
Total Marks:	25 (5 + 20)
Exam Duration:	45 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (90) out of 895 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Select Type Award: 1 Penalty: 0 Theory of Computation

Which of the following languages are undecidable?

A. $A_1 = \{\langle M \rangle \mid M \text{ is a Turing Machine that has an even number of states } \}$
B. $A_2 = \{\langle M_1, M_2 \rangle \mid M_1, M_2 \text{ are Turing Machines and } L(M_1) \cup L(M_2) = \Sigma^*\}$
C. $A_3 = \{\langle M \rangle \mid M \text{ is a Turing Machine and } L(M) \text{ is finite } \}$
D. $A_4 = \{\langle M \rangle \mid M \text{ is an DFA } L(M) \text{ is finite } \}$

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Theory of Computation

A context-free language L is given in the form of a context-free grammar G.
Consider the following decision problems:

- $\mathrm{P1}$: Is there a push-down automaton P which recognizes L with bounded stack usage, i.e., there is a constant bound $k \in \mathbb{N}$ such that no run of P needs more than k symbols on the stack.
- $\mathrm{P2}$: Is L regular?

Which of the above decision problems are undecidable?

A. Only $\mathrm{P1}$
B. Only $\mathrm{P2}$
C. Both
D. None

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Theory of Computation

Consider the following statements :

1. Let L be a language accepted by a non-deterministic Turing machine in which every computation terminates. Then L is recursive.
2. A language L is recursive if, and only if, L and L' are recursively enumerable.

Which of the above statements is/are true?

A. Only 1
B. Only 2
C. Both
D. None

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Theory of Computation

Consider the following languages :

- $\mathrm{L1} := \{\langle \mathrm{M} \rangle \mid \mathrm{M}$ is a TM and $\mathrm{L(M)}$ is countable $\}$.
- $\mathrm{L2} := \{\langle \mathrm{M} \rangle \mid \mathrm{M}$ is a TM and $\mathrm{L(M)}$ is uncountable $\}$.

Which of the above languages is Decidable?

A. Only $\mathrm{L1}$
B. Only $\mathrm{L2}$
C. Both
D. None

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Select Type | Award: 1 | Penalty: 0 | Theory of Computation

Which of the following is/are non-trivial properties for the set of RE languages?

A. L is a countable language.
B. L is a context-free language.
C. L is a decidable language.
D. L is a Turing recognizable language.

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

Which of the following is/are true?

A. Every infinite context free language L has a subset S which is undecidable.
B. Every infinite context free language L has a subset S which is unrecognizable.
C. Every infinite context free language L has an infinite subset S which is non-regular.
D. Every infinite context free language L has an infinite subset S which is regular.

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

Which of the following statements are true?

A. If every subset of a language L is CFL, then L must be CFL.
B. If every subset of a language L is CFL, then L must be regular.
C. If every subset of a language L is decidable, then L must be decidable.
D. If every subset of a language L is decidable, then L must be finite.

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Theory of Computation

Let M be the Turing machine defined by

δ	B	a	b	c
q_0	$q_1, \mathrm{B}, \mathrm{R}$			
q_1	$q_2, \mathrm{B}, \mathrm{L}$	q_1, a, R	q_1, c, R	q_1, c, R
q_2		q_2, c, L		q_2, b, L

q_0 is the starting state. B is the blank tape symbol. Describe the result of a computation in M.

A. The result of a computation is to replace the a's in the input string with b's and the c's with b's.
B. The result of a computation is to replace the a's in the input string with c's and the c's with a's.
C. The result of a computation is to replace the a's in the input string with c's and the c's with b's.
D. The result of a computation is to replace the a's in the input string with b's and the c's with a's.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

Let set S have an infinite number of elements, and set A have a finite number of elements then which of the following statements is/are False?

A. If f is a one-one mapping from set S to set S, then f is onto.
B. If f is an onto mapping from set S to set S, then f is one-one.
C. If f is a one-one mapping from set A to set A, then f is onto.
D. If f is an onto mapping from set A to set A, then f is one-one.

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

For an automata M, let $\langle \mathrm{M} \rangle$ denote some unary encoding of M, and $\mathrm{L}(\mathrm{M})$ denotes the set of strings, over $\{0, 1\}$, accepted by M.

Which of the following languages are Turing-recognizable?

A. $\{\langle \mathrm{M} \rangle | \mathrm{M}$ is a (deterministic) Turing machine and M accepts $010\}$.
B. $\{\langle \mathrm{M} \rangle | \mathrm{M}$ is a nondeterministic Turing machine and M accepts $010\}$.
C. $\{\langle \mathrm{M} \rangle | \mathrm{M}$ is a Turing machine and M does not accept $101\}$.
D. $\{\langle \mathrm{M} \rangle \mid \mathrm{M}$ is a Turing machine and $\mathrm{L}(\mathrm{M}) = \Sigma^*\}$.

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

If A is a set, let $|\mathrm{A}|$ denotes the cardinality of set A. We say $|\mathrm{B}| \leq |\mathrm{A}|$ if and only if there exists an injection from B to A. We say $|\mathrm{B}| < |\mathrm{A}|$ if and only if there exists an injection from B to A but there is no surjection from B to A.

Which of the following is/are false?

A. Let A be an uncountable set. There is no uncountable set B with $|\mathrm{B}| < |\mathrm{A}|$.
B. Let A be a countable set. There is no infinite set B with $|\mathrm{B}| < |\mathrm{A}|$.
C. Every subset of an uncountable set is either finite or countably infinite.
D. Let L be some Non-recursive enumerable language, then L is countable.

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

Suppose there are four problems (languages) $\mathrm{A}, \mathrm{B}, \mathrm{C}$, and D. Each of these languages may or may not be recursively enumerable. However, we know the following about them:

i. There is a mapping reduction (i.e., an algorithm which can be carried out by some Turing machine, not necessarily polynomial-time) from A to B.
ii. There is a mapping reduction from B to C.
iii. There is a mapping reduction from D to C

Consider the following statements and select the statements that are CERTAIN to be true, regardless of what problems A through D are.

A. If A is recursive, then the complement of B is recursive.

B. If C is recursive, then the complement of D is recursive.
C. If C is recursively enumerable, then the union of B and D is recursively enumerable.
D. If C is recursively enumerable, then the intersection of B and D is recursively enumerable.

Your Answer: | Correct Answer: B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

For a Turing machine M, let $\langle \mathrm{M} \rangle$ denote some binary encoding of M, and $\mathrm{L}(\mathrm{M})$ denotes the set of strings, over $\{0,1\}$, accepted by M.
Which of the following languages is/are undecidable :

A. $\mathrm{L1} := \{\langle \mathrm{M} \rangle \mid \mathrm{M}$ is a TM and there exists an input on which M halts in less than $|\langle \mathrm{M} \rangle|$ steps .
B. $\mathrm{L2} := \{\langle \mathrm{M} \rangle \mid \mathrm{M}$ is a TM and $|\mathrm{L}(\mathrm{M})| \leq 3\}$.
C. $\mathrm{L3} := \{\langle \mathrm{M} \rangle \mid \mathrm{M}$ is a TM that accepts all even numbers $\}$.
D. $\mathrm{L4} := \{\langle \mathrm{M} \rangle \mid \mathrm{M}$ is a TM and $|\mathrm{L}(\mathrm{M})| \geq 3\}$.

Your Answer: | Correct Answer: B;C;D | Not Attempted | Time taken: 00min 01sec | Discuss

Q #14 | Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

Consider the following problems. $\mathrm{L}(\mathrm{G})$ denotes the language generated by a grammar G. $\mathrm{L}(\mathrm{M})$ denotes the language accepted by a machine M.

Which one of the following problems is/are decidable?

A. For an unrestricted grammar G and a string w, where $w \in \mathrm{L}(\mathrm{G})$.
B. Given a Turing Machine M, whether $\mathrm{L}(\mathrm{M})$ is regular.
C. Given two grammars G_1 and G_2 whether $\mathrm{L}\left(\mathrm{G}_1\right) = \mathrm{L}\left(\mathrm{G}_2\right)$.
D. Given an NFA N, whether there is a deterministic PDA P such that N and P accept the same language.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

Which of the following statements are true?

A. If A is a subset of B and A is uncountable, then so is B.
B. If A is uncountable and B is any set, then the union $\mathrm{A} \cup \mathrm{B}$ is also uncountable.
C. If A is uncountable and B is any set, then the Cartesian product $\mathrm{A} \times \mathrm{B}$ is also uncountable.
D. If A is infinite (even countably infinite) then the power set of A is uncountable.

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

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